भारतीय मानक ब्यूरो
(उपभोक्ता मामले, खाद्य एवं सार्वजनिक वितरण मंत्रालय, भारत सरकार)
BUREAU OF INDIAN STANDARDS
(Ministry of Consumer Affairs, Food & Public Distribution, Govt. of India

मानक: पथप्रदर्शक:

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व्यापक परिचालन मसौदा

हमारा संदर्भ : सीईडी 06/टी-16 **17 अगस्त 2026**

तकनीकी समिति : पत्थर विषय समिति, सीईडी - 06

प्राप्तकर्ता :

क) सिविल इंजीनियरी विभाग परिषद्, सीईडीसी के सभी सदस्य
ख) सीईडी 06 के सभी सदस्य
ग) रूचि रखने वाले अन्य निकाय

प्रिय महोदय/महोदया,

निम्नलिखित भारतीय मानक का मसौदा संलग्न हैं:

प्रलेख संख्या	शीर्षक
सीईडी 06 (35148)WC	**भारतीय मानक का मसौदा** **संरचनात्मक ग्रेनाइट - विशिष्टि** (IS 3316 का दूसरा पुनरीक्षण) ICS No. 91.100.15

कृपया इस मानक के मसौदे का अवलोकन करें और अपनी सम्मतियाँ यह बताते हुए भेजे कि यदि यह मानक के रूप में प्रकाशित हो तो इस पर अमल करने में आपके व्यवसाय अथवा कारोबार में क्या कठिनाइयाँ आ सकती हैं।

सम्मतियाँ भेजने की अंतिम तिथि: 01 अक्टूबर 2026

टिप्पणियाँ, यदि कोई हों, बीआईएस ई-गवर्नेंस पोर्टल https://www.services.bis.gov.in/php/BIS_2.0/dgdashboard/draft/darftdetail/63/3/CED के माध्यम से ऑनलाइन भेजी जा सकती हैं।

वैकल्पिक रूप से, टिप्पणियाँ संलग्न प्रारूप में भी दर्ज की जा सकती हैं और ced6@bis.gov.in या ced@bis.gov.in पर ईमेल की जा सकती हैं।

आपको अपनी टिप्पणियाँ प्रस्तुत करने के लिए लॉगिन करना पड़ सकता है, कृपया लॉगिन बनाएं।

यदि कोई सम्मति प्राप्त नही होती है अथवा सम्मति में केवल भाषा सम्बन्धी त्रुटि हुई तो उपरोक्त प्रलेख को यथावत अंतिम रूप दिया जाएगा। यदि सम्मित तकनीकी प्रकृति की हुई विषय समिति के अध्यक्ष के परामर्श से अथवा उनकी इच्छा पर आगे की कार्यवाही के लिए विषय समिति को भेजे जाने के बाद प्रलेख को अंतिम रूप दे दिया जाएगा ।

यह प्रलेख भारतीय मानक ब्यूरो की वैबसाइट www.bis.gov.in पर भी उपलब्ध हैं।

धन्यवाद।

भवदीय
ह/-
(श्री द्वैपायन भद्र)
वैज्ञानिक 'ई'/निदेशक एवं प्रमुख (सिविल इंजीनियरिंग)
ई–मेल: hced@bis.gov.in

संलग्न: उपरलिखित

भारतीय मानक ब्यूरो
(उपभोक्ता मामले, खाद्य एवं सार्वजनिक वितरण मंत्रालय, भारत सरकार)
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WIDE CIRCULATION DRAFT

Our Reference: CED 06/T-16 **17 August 2026**
Technical Committee: Stone Sectional Committee, CED 06

Addressed To:

a) All Members of Civil Engineering Division Council, CEDC
b) All Members of CED 06
c) All others interested

Dear Sir/Madam,

Please find enclosed the following document:

Doc No.	*Title*
CED 06 (35148)WC	**Draft Indian Standard** **Structural Granite – Specification** **(*second revision* of IS 3316)** ICS No. 91.100.15

Kindly examine the draft standard and forward your views stating any difficulties which you are likely to experience in your business or profession, if this is finally adopted as National Standard.

Last Date for comments: 01 October 2026

Comments if any, may be sent online through the BIS e-governance portal at https://www.services.bis.gov.in/php/BIS_2.0/dgdashboard/draft/darftdetail/63/3/CED .

Alternatively, comments may also be recorded in the enclosed format and emailed at ced6@bis.gov.in or at ced@bis.gov.in.

You may be required to login to submit your comments, kindly create a login.

In case no comments are received or comments received are of editorial nature, you will kindly permit us to presume your approval for the above document as finalized. However, in case of comments of technical in nature are received then it may be finalized either in consultation with the Chairman, Sectional Committee or referred to the Sectional Committee for further necessary action if so desired by the Chairman, Sectional Committee.

The document is also hosted on BIS website www.bis.gov.in.

Thanking you,

Sd/-

(**Shri Dwaipayan Bhadra**)
Scientist 'E'/Director And Head
(Civil Engineering)
E-mail: **hced@bis.gov.in**

Encl: As above

For BIS Use Only CED 06(35148)WC
August 2026

FORMAT FOR SENDING COMMENTS ON THE DOCUMENT

[Please use A4 size sheet of paper only and type within fields indicated. Comments on each clause/sub-clause/ table/figure, etc, be stated on a fresh row. Information/comments should include reasons for comments, technical references and suggestions for modified wordings of the clause. **Comments through https://www.services.bis.gov.in/php/BIS_2.0/WCDraft/comment_pdraft.php shall be appreciated**.]

Doc. No.: CED 06(35148) WC **BIS Letter Ref**: CED 06/T-16

Title: Structural Granite — Specification (*second revision* of IS 3316)

Last date of comments: **01 October 2026**

Name of the Commentator/ Organization: ____________________________

Sl No.	Clause/ Para/ Table/ Figure No. commented	Comments/ Modified Wordings	Justification of Proposed Change
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NOTE- Kindly insert more rows as necessary for each clause/table, etc

Draft Indian Standard

STRUCTURAL GRANITE — SPECIFICATION

(*Second Revision* of IS 3316)

ICS 91.100.15

Stones Sectional Committee, CED 06 Last date of comments **01 October 2026**

FOREWORD

(*Formal clause will be added later*.)

Granite is an important structural and ornamental stone and because of its high compressive strength and durability, is extensively used for massive structural works like bridge piers, sea and river walls, dams and monumental buildings, where excessive wear and abrasion is likely to occur. Fine grained variety of granite that takes and preserves high polish and is susceptible of being carved is employed for ornamental and monumental works and also for inscription purposes. Granite is found in several parts of India with slight variations in the physical properties of the stone depending upon the grain size and the minerals that go into its formation. This standard has, therefore, been formulated with a view to specifying suitable grades of granite stone and providing necessary guidance in the selection of proper grade of granite for a given specific job.

Granite is available in different colours mostly in grey, mottled grey, red, pink, dark blue, white or green depending upon the colour of the component minerals. The designer shall, therefore, specify the colour desired and the permissible natural variations in colour and texture in detail or by naming the granite having the required characteristics. Granite containing injurious minerals like pyrites and marcasite which upon exposure rust rapidly and cause objectionable strain on the stone shall be excluded. This standard was first published in 1965 and was revised in 1974 to include requirements relating to strength, dimensions marking and other details. In this revision, the provisions have been updated and rationalized in line with the latest relevant Indian Standards. provisions relating to the requirements, dimensions, marking and sampling have also been reviewed and suitably modified, wherever necessary. A new table specifying the physical properties of structural granite has also been included.

This standard contributes to the United Nations Sustainable Development Goal 9 'Industry, innovation and infrastructure'; and Goal 12 'Ensure sustainable consumption and production patterns' towards substantial reduction of waste generation through prevention, reduction, recycling and reuse.

For the purpose of deciding whether a particular requirement of this standard is complied with, the final value, observed or calculated, expressing the result of a measurement shall be rounded off in accordance with IS 2 : 2022 'Rules for rounding off numerical values (second revision)'. The number of significant places retained in the rounded off value should be the same as that of the specified value in this standard.

Draft Indian Standard

STRUCTURAL GRANITE — SPECIFICATION

(*First Revision* of IS 3316)

1 SCOPE

This standard covers the selection, grading and strength requirements of structural granite for the various constructional uses.

2 REFERENCES

The Indian Standards listed in **Annex A** contain provisions, which through reference in this text, constitute provisions of this standard. At the time of publication, the editions indicated were valid. All standards are subject to revision, and parties to agreements based on this standard are encouraged to investigate the possibility of applying the most recent edition of these standards.

3 GENERAL REQUIREMENTS

Granite shall be free from flaws, injurious veins, cavities and similar imperfections that would impair its structural integrity and would adversely affect its strength and appearance.

4 STRENGTH AND DURABILITY REQUIREMENTS

4.1. Sample and test requirements. The sample size and corresponding test requirements are specified in Table 1. It is to be noted that, for the strength requirements, specimen (having height > 0.9 times width) shall be tested for both compressive and transverse strength, and slabs shall be tested for transverse strength only.

4.2 The procedure and evaluation for carrying out the transverse strength test are based on **5** and **4**, respectively, in IS 1121(Part 2).

4.3 The apparent specific gravity and water absorption tests shall be carried out on a test specimens weighing about 1 kg.

TABLE 1 PHYSICAL PROPERTIES OF STRUCTURAL GRANITE

Sr. No	Characteristics	Requirements	Method of Test Reference to Indian Standards
1.	Compressive Strength, in N/mm^2, *Min*.	100	IS 1121 Part 1
2.	Transverse strength, in N/mm^2, *Min*.	10	IS 1121 Part 2

4.	Apparent Specific Gravity, *Min*.	2.6	IS 1124
5.	Maximum Water Absorption, in percent, *Max*	0.5 %	IS 1124
6.	Abrasion Resistance Test, in mm, *Max*	10	IS 1706
7.	Linear Thermal Expansion, in first decimal x $10^{-6}/^{\circ}$ C	10	IS 13630 Part 4

5 DIMENSIONS

5.1 Slabs – The slabs shall be rectangular or square in shape and of specified dimensions. The tolerance in length and breadth shall be ± 2 mm and thickness ± 1 mm. The bottom face shall be dressed back square with the top surface for at least 50 mm, without hollowness or spalling off.

5.2 Blocks for Masonry – The dimensions of the blocks shall have at least one dimension more than 300 x 200 x 100 mm. The tolerance shall be allowed ± 5 mm for facing blocks. The edges of the blocks shall be dressed according to IS 1129.

6 MARKING

6.1 Each type of blocks and slabs may be marked in a suitable manner with the manufacturer's identification mark or initials.

6.2 BIS Certification Marking

The product(s) may be marked with Standard Mark as per the conformity assessment schemes governed by the provisions of the Bureau of Indian Standards Act, 2016 and the Rules and Regulations made there under. The details of conditions for the license may be obtained from the Bureau of Indian Standards.

7 SAMPLING

7.1 Lot – In any consignment all the blocks/slabs of the same quarry shall be grouped together to constitute a lot.

7.1.1 Samples shall be selected and tested separately for each lot for determining its conformity or otherwise to the requirements of the specification.

7.2 The number of blocks/slabs to be selected for the sample shall depend upon the size of the lot and shall be in accordance with Table 1.

TABLE 2 SAMPLE SIZE AND CRITERIA FOR CONFORMITY

NUMBER OF BLOCKS/ SLABS IN THE LOT	NUMBER OF BLOCKS/ SLABS TO BE SELECTED IN SAMPLE	PERMISSIBLE NUMBER OF DEFECTIVES	SUB-SAMPLE SIZE IN NUMBER
(1)	**(2)**	**(3)**	**(4)**
Up to 100	5	0	3
101 to 300	8	0	3
301 to 500	13	0	6
501 to 1 000	20	1	6

7.2.1 The blocks/slabs in the sample shall be selected at random and in order to ensure the randomness of selection, random number tables may be used (*see* IS 4905).

7.3 All the blocks/slabs selected in accordance with col 2 of Table 2 shall be examined for general requirements and dimensions (*see* **3** and **4**). Any block/slab failing in any one or more of the above requirements shall be considered as defective. A lot shall be considered as conforming to those requirements if the number of defective blocks/slabs obtained is not more than the permissible number of defectives given in col 3 of Table 2.

7.4 The lot having been found satisfactory with respect to dimensions, and general requirements shall be tested for physical properties. For this purpose a sub-sample of size given in col 4 of Table 2 shall be selected at random. These blocks/slabs in the sub-sample shall be first tested for compression strength, the same samples after completion shall be utilized for determining the other properties as per Table 1 (*see* **4**). A lot shall be considered having been satisfied the requirements of the physical properties if none of the blocks/slabs tested for these requirements fails in any of these tests

ANNEX A
(*Clause* 2)
LIST OF REFERRED STANDARDS

IS No.	*Title*
IS 1121	Determination of Strength Properties of Natural Building Stones — Methods of Test
Part 1: 2023	Part 1 Compressive Strength (*third revision*).
Part 2: 2023	Part 2 Transverse Strength (*third revision*).
IS 1124 : 1974	Method of test for determination of water absorption and apparent specific gravity of natural building stones (*first revision*).
IS 1129 : 1972	Recommendations for dressing of natural building stones (*first revision*).
IS 1706 : 2024	Method for determination of resistance to wear by abrasion of natural building stones (*second revision*)
IS 4905 : 2015	Random sampling and randomization procedures (*first revision*)
IS 13630 Part 4 : 2019	Ceramic Tiles - Methods of Test, Sampling and Basis for Acceptance Part 4 Determination of Linear Thermal Expansion (*Second Revision*)